

The Role of Optics in Next-Generation Multi-Gigawatt Datacenters

B. Gangopadhyay⁽¹⁾, D. Mukherjee⁽²⁾, A. Vakili⁽³⁾ J. Rahn⁽³⁾

- (1) Meta Platforms Inc., Network Infrastructure, United Kingdom, (bgangopadhyay@meta.com)
(2) Meta Platforms Inc., Network Infrastructure, Washington DC, USA
(3) Meta Platforms Inc., Network Infrastructure, Menlo Park, USA

Abstract Multi-gigawatt AI datacenters require optical interconnect systems that scale from intra-cluster fabrics to inter-building backbone rings while meeting stringent power, latency, and reliability constraints. We describe the optical architecture deployed across disaggregated AI fabrics and metro-ring backbone networks, detailing technology choices including 2×400G FR4-LITE intra-datacenter optics, 800GZR coherent backbone modules, and optical protection switching for high availability. ©2026 The Author(s))

Introduction

Distributed training of foundation models demands sustained all-to-all communication among hundreds of thousands of accelerator chips over lossless RoCE fabrics with microsecond-scale tail latency [1, 2]. Hyperscale infrastructure investment has correspondingly accelerated, with planned multi-gigawatt datacenter complexes where individual clusters surpass one gigawatt of IT load [3, 4]. These facilities span multiple buildings across campus-scale sites, requiring optical connectivity at every level of the network hierarchy.

The optical interconnect challenge is twofold. Within a cluster, hundreds of thousands of 800G links must deliver low-latency, lossless connectivity between accelerator racks and spine switches across distances from metres to two kilometres. Between buildings and across the campus backbone, DWDM ring networks must provide massive aggregate capacity with rapid optical-layer protection switching—all while operating within tight power envelopes where every watt consumed by the network competes with compute. A 51.2 Tbps switch populated with 64 × 800G transceivers consumes approximately 960 W in optics alone [5, 6], and at campus scale the aggregate optical power

reaches tens of megawatts. Fig. 1 illustrates the end-to-end optical architecture spanning these domains.

Intra-Cluster Fabric and Optical Layer

The AI training clusters described herein employ two complementary fabric architectures, both presented at the Open Compute Project (OCP) Summit. The Disaggregated Fabric (DSF), introduced at OCP 2024 and expanded with a dual-stage non-blocking design in 2025, achieves deterministic lossless delivery through centrally coordinated time-slot allocation over standard Ethernet switches [7, 8]. For the largest clusters such as Prometheus (exceeding one gigawatt), the Non-Scheduled Fabric (NSF) uses shallow-buffered, disaggregated Ethernet switches optimised for scale-out topologies [8].

Both architectures run the open-source FBOSS network operating system with the OCP Switch Abstraction Interface (SAI), enabling multi-vendor silicon support. The switching platforms—Minipack3 (Broadcom Tomahawk5) and Minipack3N (NVIDIA Spectrum-4)—deliver 51.2 Tbps across 64 retimer-free OSFP ports [9]. The retimer-free design is a deliberate optical-aware system choice: by eliminating retimers, each switch saves significant power and latency at the cost of tighter signal integrity requirements on the optical link.

The intra-datacenter optical module selection follows a tiered strategy [8, 9]: 2×400G FR4-LITE modules provide 800G connectivity for reaches up to 500 metres, optimised for intra-building links where shorter reach enables lower cost and power. For longer intra-campus spans up to 3 km, 2×400G FR4-BASE optics are deployed. Additionally, 400G/2×400G DR4 modules serve point-to-point requirements. This tiered approach matches optical technology to distance requirements, avoiding over-engineering shorter links while ensuring adequate performance margin on longer reaches.

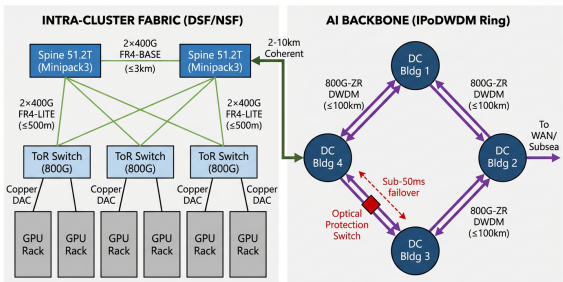


Fig. 1: End-to-end optical architecture for multi-gigawatt AI datacenters.

AI Backbone: Ring Topology and Optical Protection

The backbone network interconnecting datacenter buildings required a 10× capacity increase, with the deployment timeline accelerated from 2028 to 2024–2025 to match the pace of AI workload growth [10, 11]. Three key techniques enabled this rapid scaling: pre-building scalable datacenter metro architectures with optical fiber ring topologies, vertical platform scaling (higher per-port rates and higher port counts), and horizontal scaling (additional parallel systems) [11].

The backbone employs an IP-over-DWDM (IPoDWDM) architecture with 800ZR and 800ZR+ coherent pluggable transceivers integrated directly into routing platforms [12]. This eliminates separate optical transport equipment, reducing both capital cost and operational complexity. Each backbone node handles 64 × 800 Gbps (51.2 Tbps), with connections spanning campus-scale distances without intermediate amplification [11].

A defining architectural choice is optical protection switching at the DWDM ring layer. Rather than relying on IP-layer rerouting (which requires duplicate router interfaces), optical protection switches traffic at the physical layer with millisecond-class failover times—significantly reducing the number of IP interfaces required on backbone routers [11]. This yields substantial power savings: fewer router line cards means less silicon and less cooling, whilst the optical protection switch itself consumes negligible power compared to equivalent electrical capacity. At multi-gigawatt campus scale with dozens of inter-building ring seg-

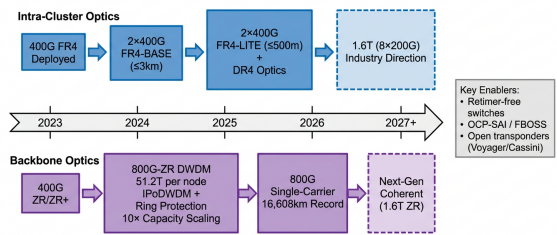


Fig. 2: Optical technology deployment roadmap

ments, this architectural choice yields substantial router power savings at scale.

The maturity and reach of coherent technology continue to advance. Recent field trials have demonstrated real-time 800 Gb/s single-carrier transmission across 16,608 km of unregenerated transpacific submarine cable—establishing a distance record for single-carrier 800G [13]. While these demonstrations employed purpose-built transponder line systems rather than pluggable ZR modules, they validate the fundamental performance headroom of 800G

coherent technology and confirm that campus-scale deployments operate well within technology margins.

The OCP-originated open packet transponder platform Cassini pioneered this disaggregated backbone optical approach, enabling operators to deploy coherent DWDM capacity on commodity white-box hardware with open network operating systems [14]. Fig. 2 shows the technology roadmap spanning these developments.

Power Considerations and Industry Outlook

The optical power challenge intensifies with each generation. Current 800G DSP-based modules at 12–18 W per port represent a significant fraction of switch-level power [5, 6]. The selection of retimer-free platforms combined with reach-optimised modules (FR4-LITE for ≤500 m rather than full FR4 for ≤2 km) exemplifies system-level power co-optimisation—matching optical module capability precisely to link distance requirements.

Several next-generation efficiency approaches are under active development across the industry for scale-out and front-end network interconnects. Linear Pluggable Optics (LPO), formalised through a multi-source agreement in 2024, or Linear Retimed Optics (LRO) delivers power savings of up to ~ 50% by power savings [15]. Co-Packaged Optics (CPO) integrates

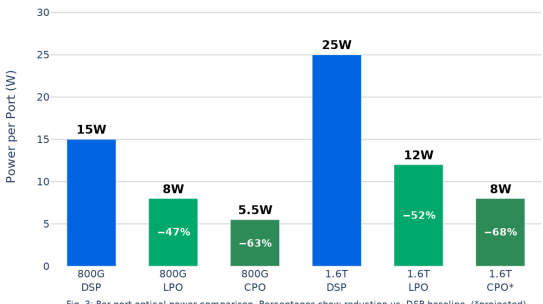


Fig. 3: Per-port optical power comparison. Percentages show reduction vs. DSP baseline (*projected)

photonic engines adjacent to the switching silicon further reducing the power allocated to support the ASIC to optics electrical interface,, achieving approximately 5.5 W per 800G port in published demonstrations—a 3× improvement [16]. These represent potential future paths as the industry scales toward 1.6T (8×200G lanes, targeting 2026–2027 volume production) and beyond [17].

Fig. 3 compares per-port power across current and emerging approaches. At facility scale with hundreds of thousands of ports, even modest per-port reductions translate to megawatts of savings—power directly reallocatable to additional compute capacity.

Scaling to Gigawatt Clusters: Prometheus and Beyond

The optical architecture described above is scaling to meet the demands of the largest AI training deployments. DSF supports non-blocking interconnection of up to 18,432 XPU (AI accelerators) per cluster at datacenter-building scale, with the dual-stage design enabling AI training across entire buildings [8, 9]. The planned deployment trajectory targets 1 GW of compute capacity during 2025, with over 1.3 million GPUs in aggregate across the infrastructure by year-end [3].

The Prometheus cluster—the first infrastructure designed to exceed 1 GW—targets launch in 2026, utilising the NSF architecture with Minipack3/3N switches and the full tiered optical portfolio described in this paper [4, 8]. Hyperion, located in Louisiana, targets over 2 GW by 2030 with architectural provisions for scaling to 5 GW, employing liquid cooling and high-performance optical networking across facilities spanning hundreds of hectares [4]. Additional multi-gigawatt clusters are in various stages of planning and construction.

At 1.3 million GPUs interconnected via 51.2 Tbps switches, the total optical transceiver count across Meta's infrastructure reaches into the millions—each requiring careful technology selection balancing reach, power, and cost. The adoption of Ethernet for both scale-out and scale-up networking further intensifies optical bandwidth demand per accelerator, as every GPU-to-GPU communication path traverses the optical fabric [9].

Conclusions

The optical interconnect layer in multi-gigawatt AI datacenters operates across two distinct but equally critical domains: intra-cluster fabrics demanding high port density with reach-optimised pluggable modules, and campus backbone rings requiring coherent DWDM with optical protection for capacity and resilience. The architecture presented in this work demonstrates how co-optimisation of the optical layer with fabric design—retimer-free switches, tiered reach modules (FR4-LITE/FR4-BASE/DR4), IPoDWDM backbone with optical ring protection—delivers power-efficient scaling at unprecedented facility scales. As the industry progresses toward 1.6T transceivers and beyond, continued innovation in optical integration and standardisation through open initiatives such as OCP will remain essential to sustaining the pace of AI infrastructure scaling.

Acknowledgements

The authors acknowledge contributions from colleagues across the Network Infrastructure and AI Platform teams at Meta, and the broader OCP community.

References

- [1] J. Dean et al., "Large Scale Distributed Deep Networks," *Proc. NeurIPS*, vol. 25, pp. 1223–1231, 2012..
- [2] A. Mittal et al., "Revisiting Network Support for RDMA," *Proc. ACM SIGCOMM*, pp. 313–326, 2018. DOI: 10.1145/3230543.3230557
- [3] M. Zuckerberg, "Building the Future of AI at Meta," *Meta Newsroom*, Jan. 2025.
- [4] "Meta to invest hundreds of billions in multi-gigawatt AI data centers," *SiliconANGLE*, Jul. 2025
- [5] Lightcounting, "800G Optical Transceivers: Market and Technology Analysis," 2024.
- [6] 800G Pluggable MSA, "800G Module Specifications," 2024. Available: <https://800gmsa.com/>
- [7] A. Saidi et al., "Disaggregated Scheduled Fabric," *Proc. OCP Global Summit*, San Jose, USA, 2024
- [8] "OCP 2025 Meta Keynote: Scaling AI Infrastructure to Data Center Regions," *IEEE ComSoc Technology Blog*, Oct. 2025.
- [9] "Meta Advances Open Network Fabrics for AI Clusters at OCP Summit 2025," *Converge Digest*, Oct. 2025.
- [10] "10x Backbone: Scaling Backbone Connectivity to Serve AI Demands," *@Scale Conference*, 2025.
- [11] "Meta: Scaling Network Infrastructure to Support AI Workload Growth at Hyperscale," *@Scale Networking*, 2025.
- [12] "Cloud and DCI Creates a New Coherent Optics Wave," *Dell'Oro Group Report*, 2025
- [13] "Ciena and Meta Achieve World Record 800 Gb/s Across 16,608 km," *Ciena Press Release*, 2024
- [14] "Cassini Open Packet Transponder," *OCP Contribution*, Edgecore Networks
- [15] "LPO MSA Specification v1.0," *LPO Multi-Source Agreement*, 2024. Available: <https://www.lpomsa.org/>
- [16] Broadcom, "Co-Packaged Optics for Hyperscale Data Centers," *OCP Contribution*, 2024
- [17] OIF, "CEI-224G Interface Specification for 200G/lane," 2025. Available: <https://www.oiforum.com/>